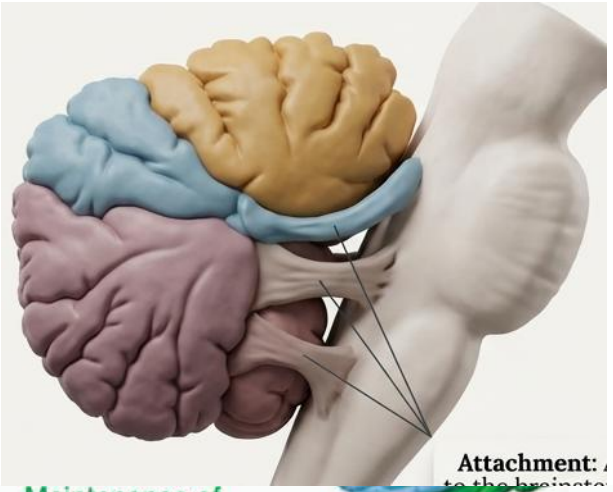
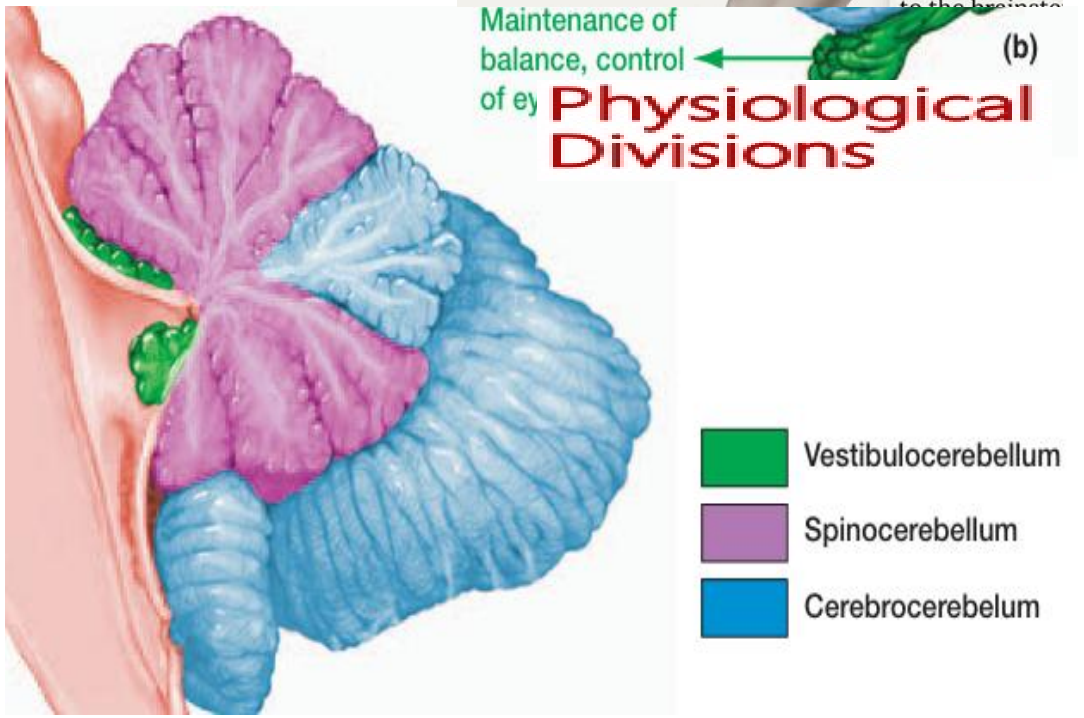
LOS;

- Cerebellum
- Functions & Dysfunction Of Cerebellum.

Macro architecture;

- Largest part of hind brain,
- Attached to brain stem by three peduncles
- located in posterior cranial fossa.
- two cerebellar hemispheres joined by vermis.





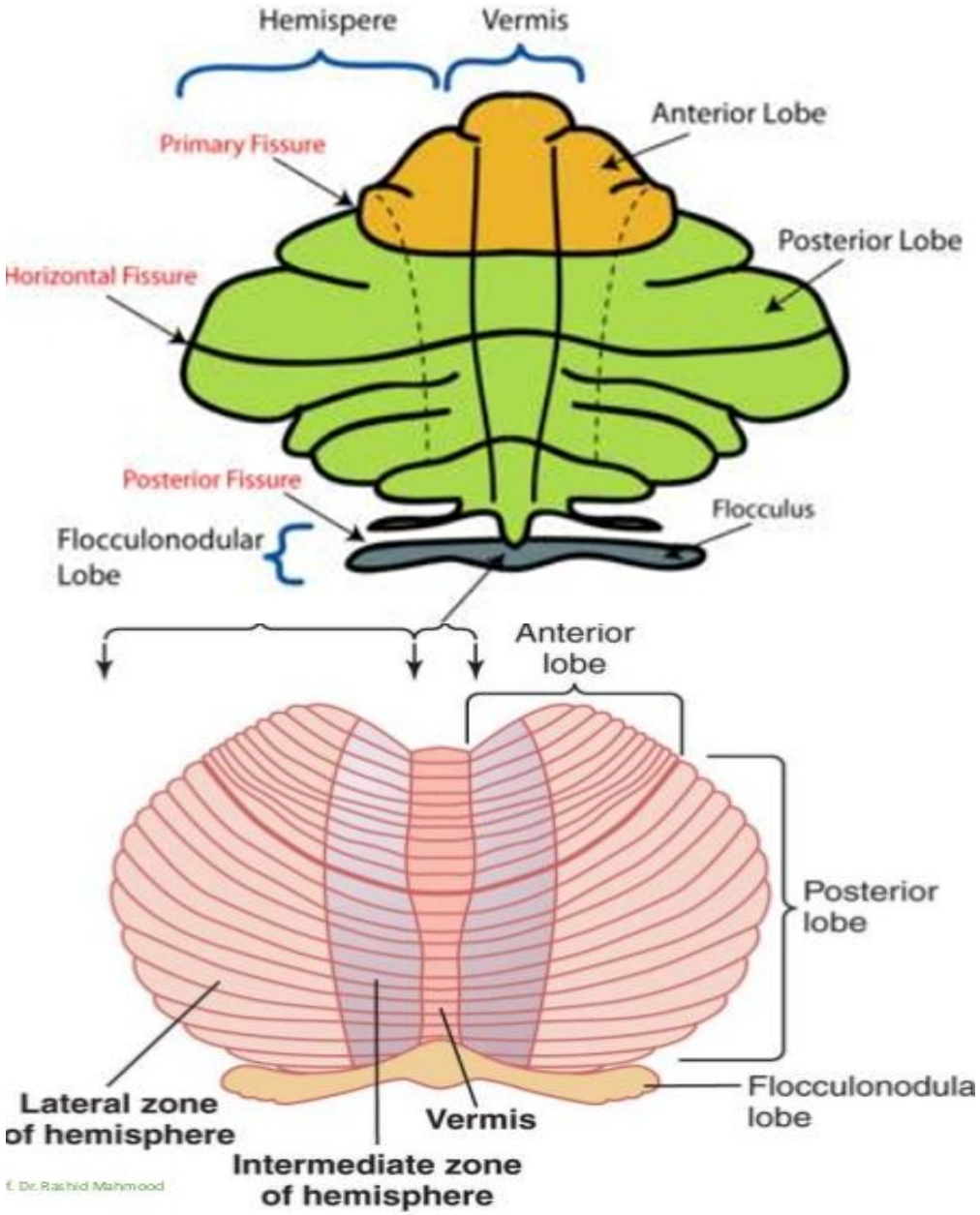
Lobes of cerebellum

Anatomically;

- **Three Lobes**

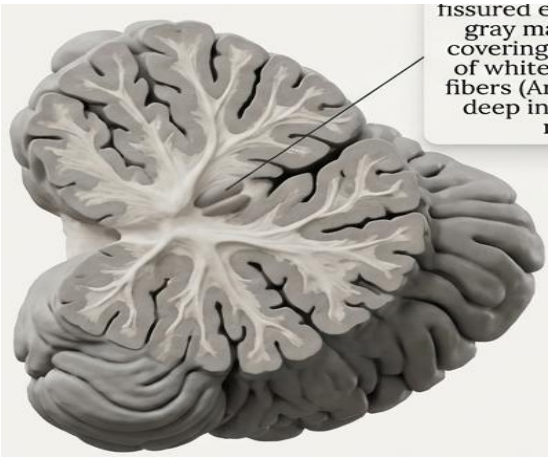
Physiologically;

- **3 zones**



Micro composition;

Consists of Highly folded and fissured white matter and a thin layer of gray matter i.e. cerebellar cortex.

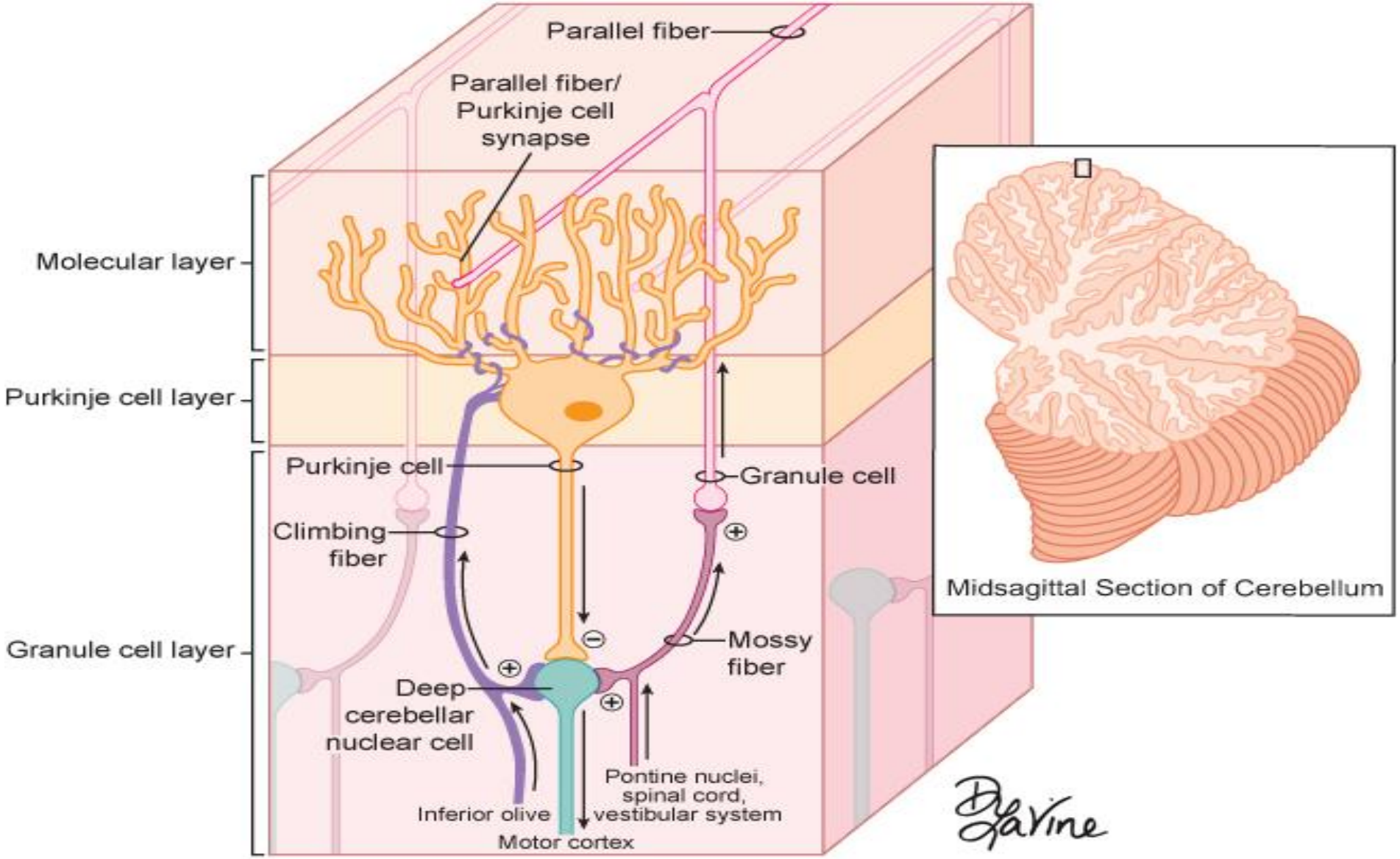


Cerebellar cortex

1. Purkinje cells
2. Granule cells
3. Stellate cells and basket cells
4. Golgi cells

Intracerebellar nuclei

1. Nucleus fastigious
2. Globose
3. Emboliform
4. Dentate nucleus


Parallel fiber
Parallel fiber/
Purkinje cell
synapse
Molecular layer
Purkinje cell layer
Purkinje cell
Granule cell
Climbing
fiber
Granule cell layer
Mossy
fiber
Deep
cerebellar
nuclear cell
Pontine nuclei,
spinal cord,
vestibular system
Inferior olive
Motor cortex
Midsagittal Section of Cerebellum

Afferent pathways from other parts of brain

Corticocerebellar; motor, premotor and somatosensory cortex

- to lateral divisions of cerebellar hemisphere

brain stem–

- Olivo-cerebellar, vestibulo-cerebellar and reticulo-cerebellar

periphery include

- Anterior & posterior spino-cerebellar, Cuneo-cerebellar pathways

Afferent nerve fibers

I Climbing fibers—derived from inferior olivary nuclei (excitatory)

- **Synapse with** dendrites of **purkinje** cells & **deep cerebellar nuclei**

II Mossy fibers—excitatory, represent **majority of input** to cerebellum

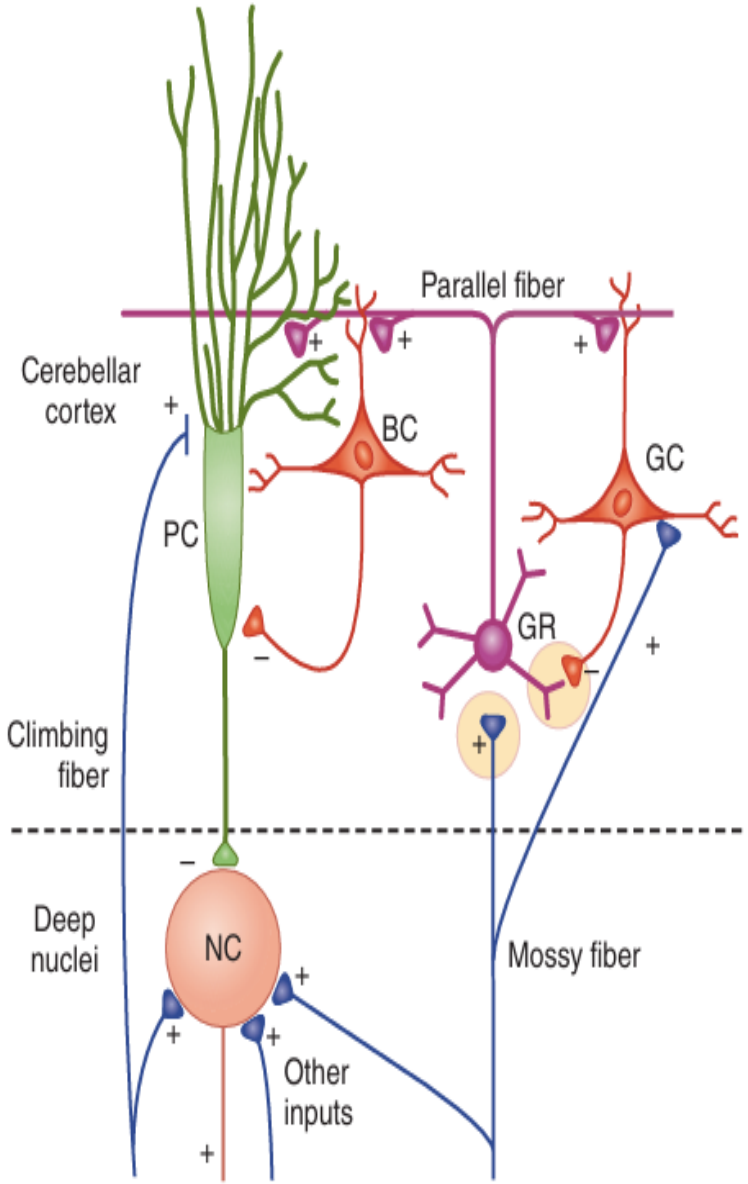
- **proprioceptive inputs from all parts of body**
- **inputs from cerebral cortex**

Synapse with

- **granule cells**& **deep cerebellar nuclei**

Efferent fibers

- Axons of Purkinje cell form **the only efferent output from cerebellar cortex (inhibitory) synapse with deep cerebellar nuclei**
- The deep cerebellar nuclei send **excitatory** fibers **to other portions of nervous system**



Excitatory input from Climbing & mossy fibers on

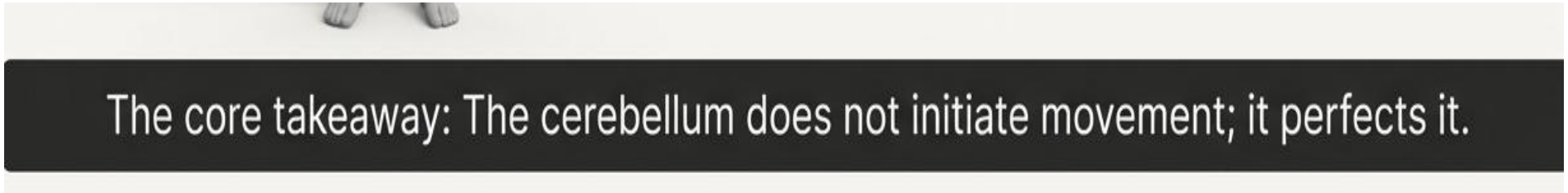
- **Excitatory cells** (granule &, basket cells)
- **Excites the Purkinji cells** that inhibit the deep nuclei.

Deep nuclei are

- **Directly Excited by Climbing And Mossy Fibers.**
- **inhibited by Purkinji fibers**
- so, an alternating **excitatory and inhibitory discharge from deep nuclei to Agonists and Antagonists occurs.**

Functions of cerebellum in overall motor control

Core Function: The highest center for the automatic regulation of reciprocal innervation between Agonistic and antagonistic muscles



Evolutionary zone Vestibulo-cerebellum/ Archicerebellum

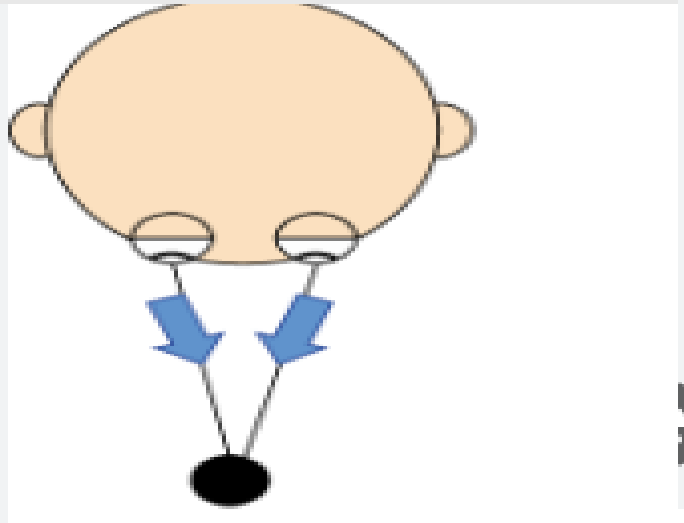
flocculo-nodular lobe and adjacent vermis

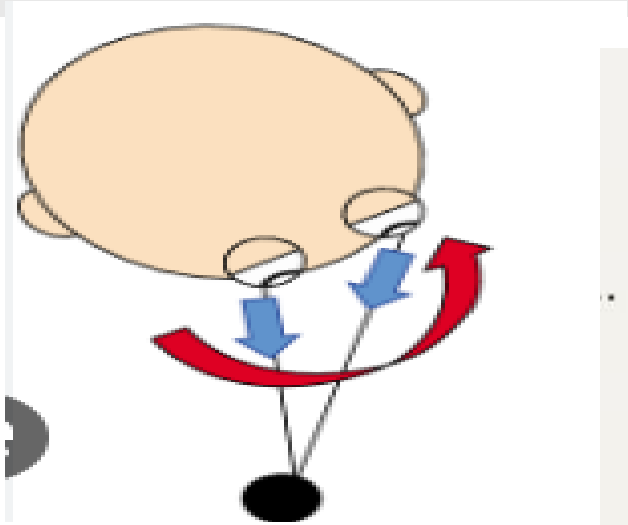
Functions:

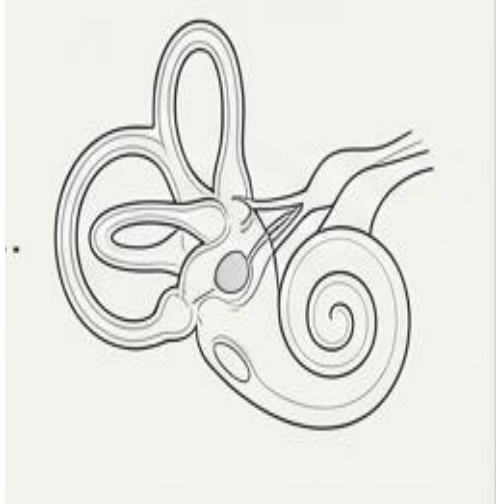
- 1.**Controls body equilibrium**
- maintains posture and equilibrium –(static and during performance of rapid movement with changing direction)
- Lesions: involved in motion sickness. produce trunk ataxia; inability to maintain balance on standing & walking as the space loses its spatial horizon.

- 2. **Facilitates** muscle tone and stretch reflex.
- 3. **Governs** the **vestibulo-ocular reflex (VOR)** to maintain visual fixation during angular acceleration and deceleration

  Lesion; Produces Nystagmus

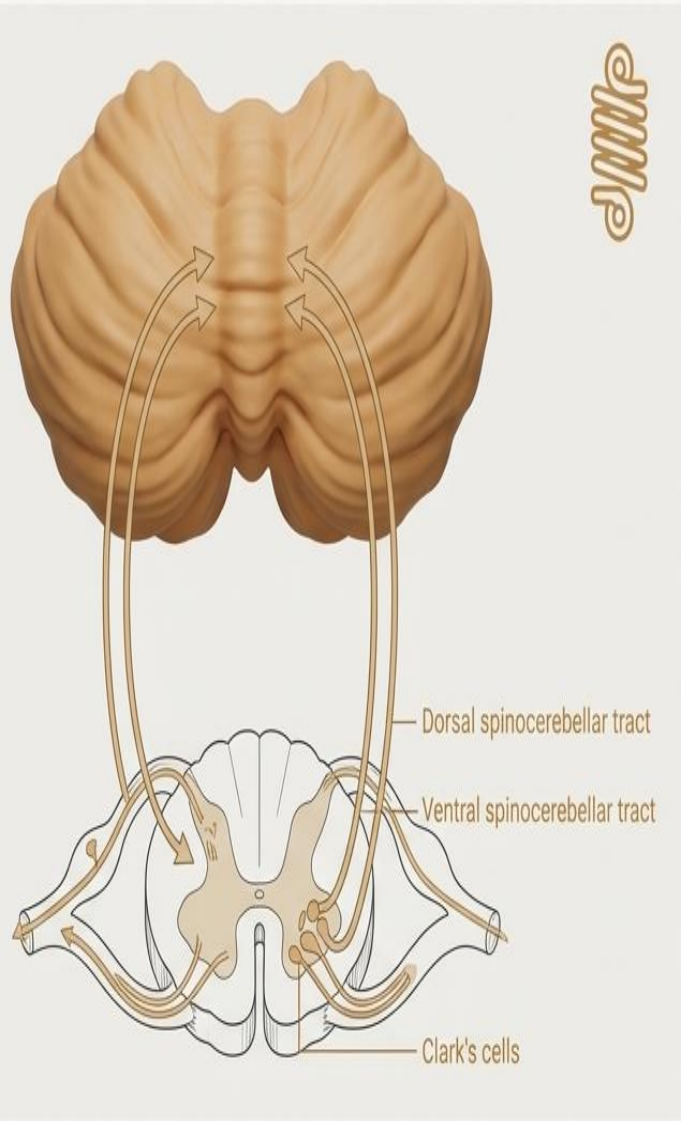






Spinocerebellum / Paleocerebellum

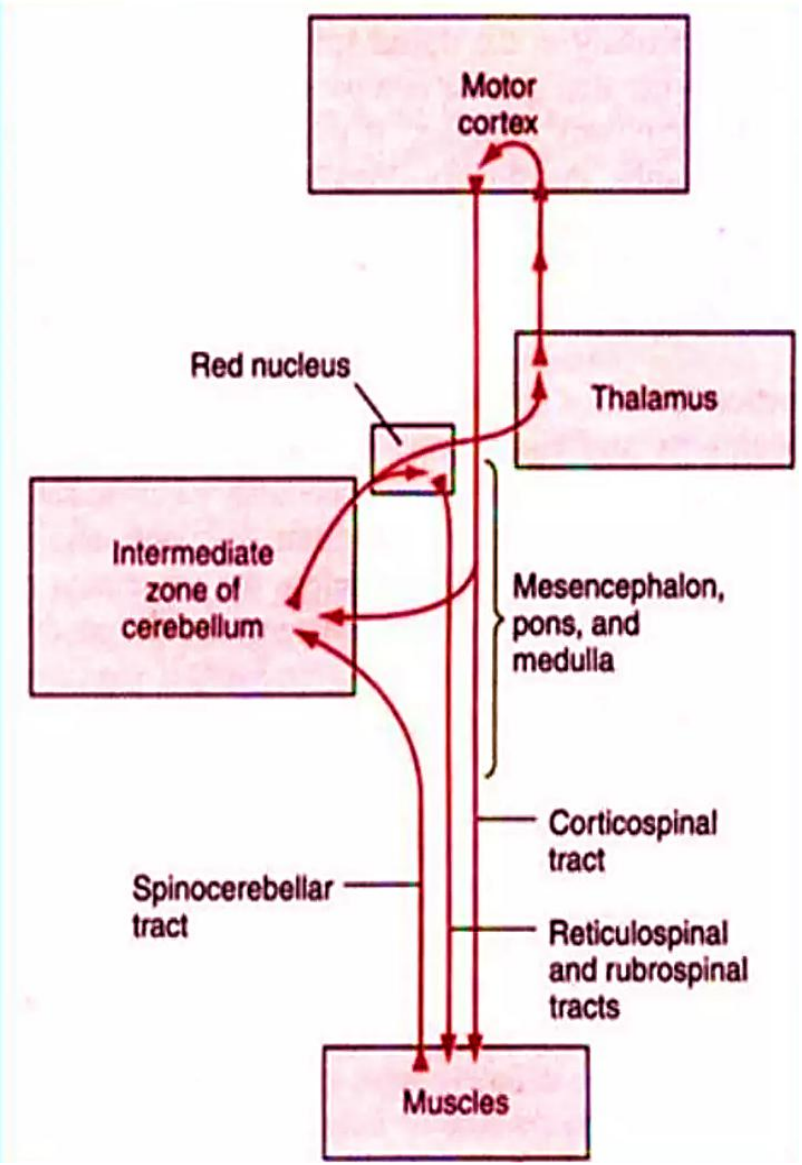
1. **Regulates muscle tone and posture** across the eyeballs, head, trunk and limbs via alpha and gamma motor neurons.
2. **Compares** the **plan** of movement with actual **performance** of movement **to correct** any discrepancy.
3. **Controls** the rate, range and direction of movement
4. **Regulates** the force of the movement.



Mechanism (Comparator)

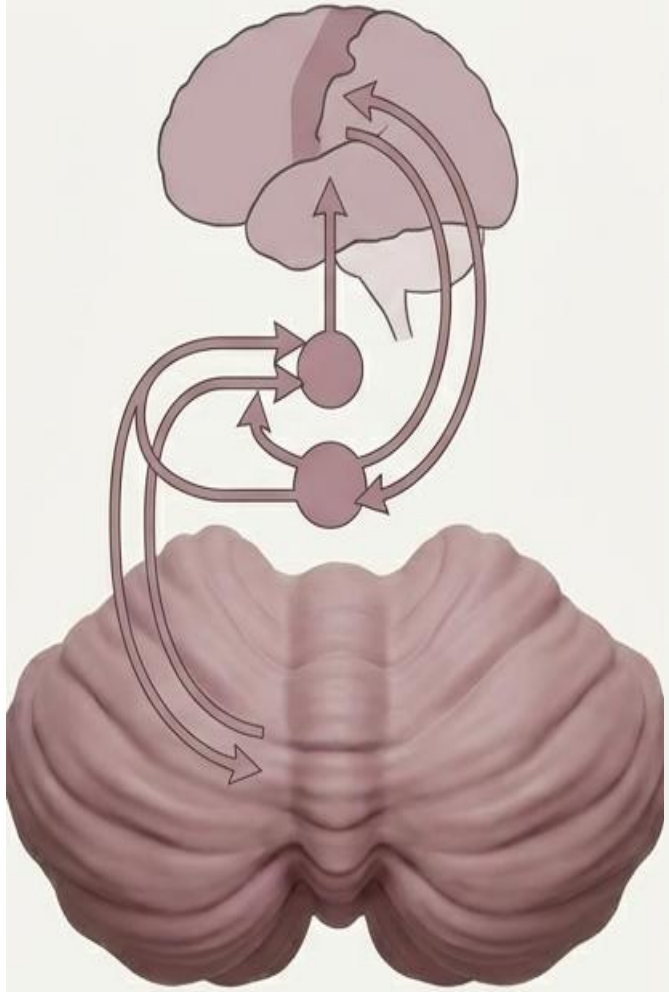
- Receives two types of information;

1. **Order from cerebral cortex** and red nucleus– about intended plan of movement
2. **Feedback from spinocerebellar tracts** from peripheral parts of body: distal proprioceptors of limb, tactile, visual and auditory receptors- **to tell cerebellum what actual movement is occurring**



- 5. **Provides** smooth coordinated movements of agonist and antagonists for performance of purposeful patterned movement. During movement, agonists contract and antagonists are inhibited (smooth coordination between agonist and antagonists).
- 6… reverse happens- to **Dampen** the movement to prevent overshooting at the end of movement.
- 7…**Controls** ballistic movements– such as typing, cycling, playing piano, saccadic movement of eyes while reading

Cerebro-cerebellum /Neocerebellum/ the processor



Evolutionary Zone

Neocerebellum (Cerebrocerebellum)

Primary Role

Works intimately with the cerebral cortex to plan and execute rapid, complex voluntary movements (e.g., typing, playing piano, talking).

Mechanism

Assesses the rate of movement and exerts a powerful negative feedback “braking effect” to stop motions precisely at the intended target.

- Concerned with **programming and planning of sequential movement**
- **Mainly connected to cerebral cortex**

1. Controls the **timing of different components of movements**

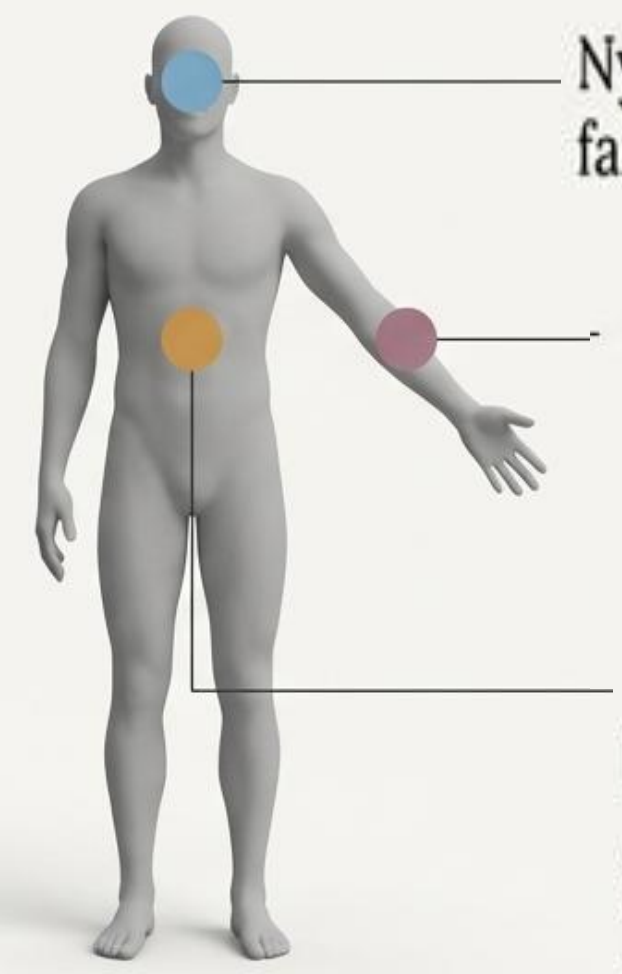
2. Smooth progression of one movement to other

3. Also has a **predictive function** from changing visual scenes and auditory phenomenon about how rapidly a person is approaching an object

The Master Functional Matrix

Anatomical Lobe	Evolutionary Zone	Key Function	Primary Input	Classic Lesion Sign
Flocculonodular	Archicerebellum	Equilibrium & VOR	Vestibular System	Trunk Ataxia
Anterior	Paleocerebellum	Posture & Muscle Tone	Spinal Cord	Hypotonia
Posterior	Neocerebellum	Planning & Braking	Cerebral Cortex	Dysmetria / Intention Tremor

Clinical Manifestations of System Failure



Ocular Deficits

Nystagmus and spontaneous deviation caused by a failure of the Flocculonodular vestibulo-ocular reflex.

Disturbances of Movement

Dysmetria (overshooting targets), Dysdiadochokinesia (inability to perform rapid alternating movements), and general decomposition of fluid movement due to a loss of Posterior braking.

Disturbances of Posture

Hypotonia and unsteady attitudes resulting from a loss of Anterior lobe muscle tone regulation.

Cerebellar disease /Lesion of Deep Cerebellar Nuclei

motor disorders without paralysis or sensory loss

1. **HYPOTONIA lack of facilitatory effect on gamma efferent discharge.**

2. **CEREBELLAR ATAXIA Lack of coordination of movements in a voluntary activity** Due to errors in rate, range, force and direction of movement.

3. **REBOUND PHENOMENON** Inability to stop movement due to incoordination **Pendular Knee jerk.**

4. ATTITUDE

- **Face** is rotated **opposite to the affected side**
- **Homologous shoulder** is **raised** and **in front of the opposite**
- **Leg** is **adducted and externally rotated**
- **Asthenia…** Feebleness: **weak grasp** and **tendency to fall to affected side---**

5. ASYNERGIA

Disorder of movement in term of **range, direction and force**.

Due to imperfect employment of synergistic components of a movements **due to lack of co-ordination**

6. DECOMPOSITION of movement

single smooth movement is broken into components

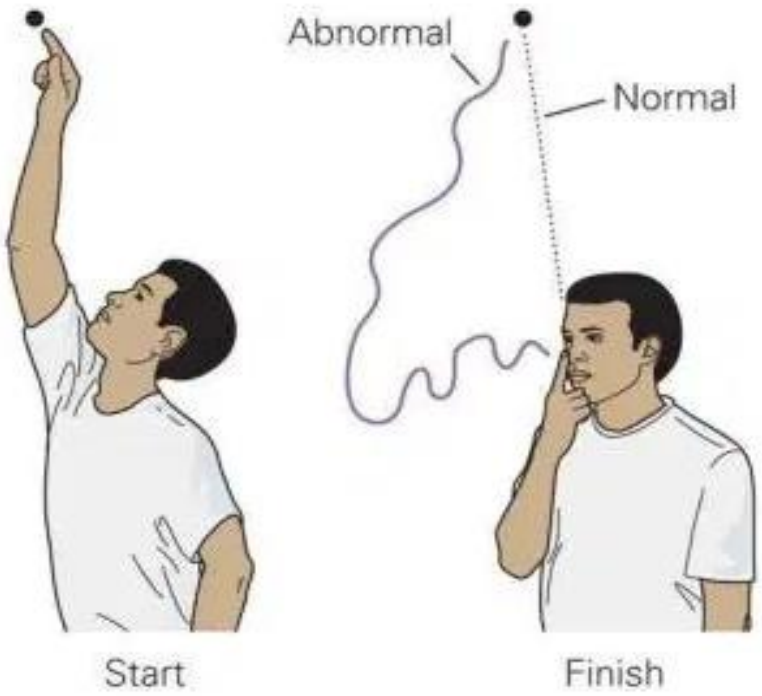
7. DYSDIADOCHOKINESIA

Inability to perform **rapid, alternate movements**

.

8.DYSMETRIA

- **Loss of sense of distance---**
- loss of ability to anticipate (**predict)** how far the movement will go
- Results in **under shooting** and **overshooting**



9. INTENSION TREMORS

Not present at rest, Appear during voluntary activity due to dysmetria.

10. GAIT: Patient walks on broad base (ataxia)

d**runken/ staggering gait**

11. SPEECH

Scanning or slurred

due lack of co-ordination

of muscle of articulation

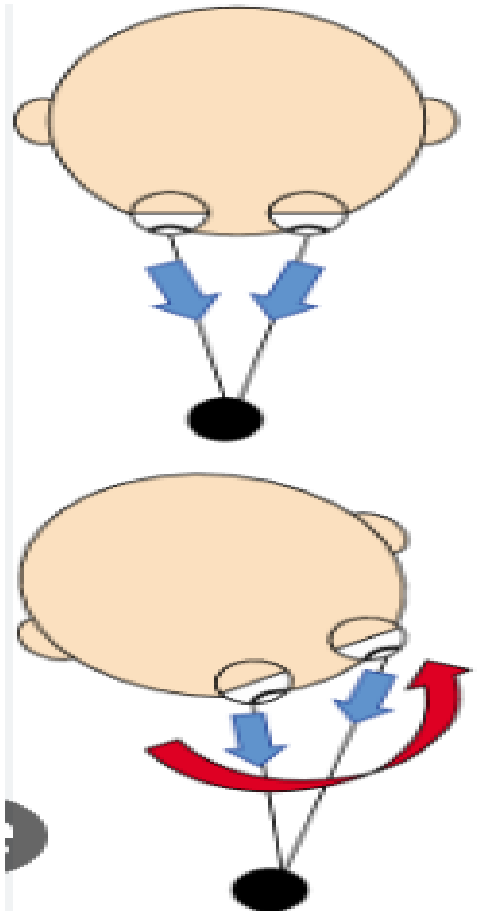
12. DYSGRAPHIA

Written speech—

To keep her balance the child with ataxia walks bent forward with feet wide apart. She takes irregular steps, like a sailor on a rough sea or someone who is drunk.

14. NYSTAGMUS

- Jerky movements of eyes (dancing eyes) with a slow and a fast component
- Represent intension tremor of eyes

Slow component---is towards the side of lesion

Fast components---is opposite to side of lesion, corrective component